

Module

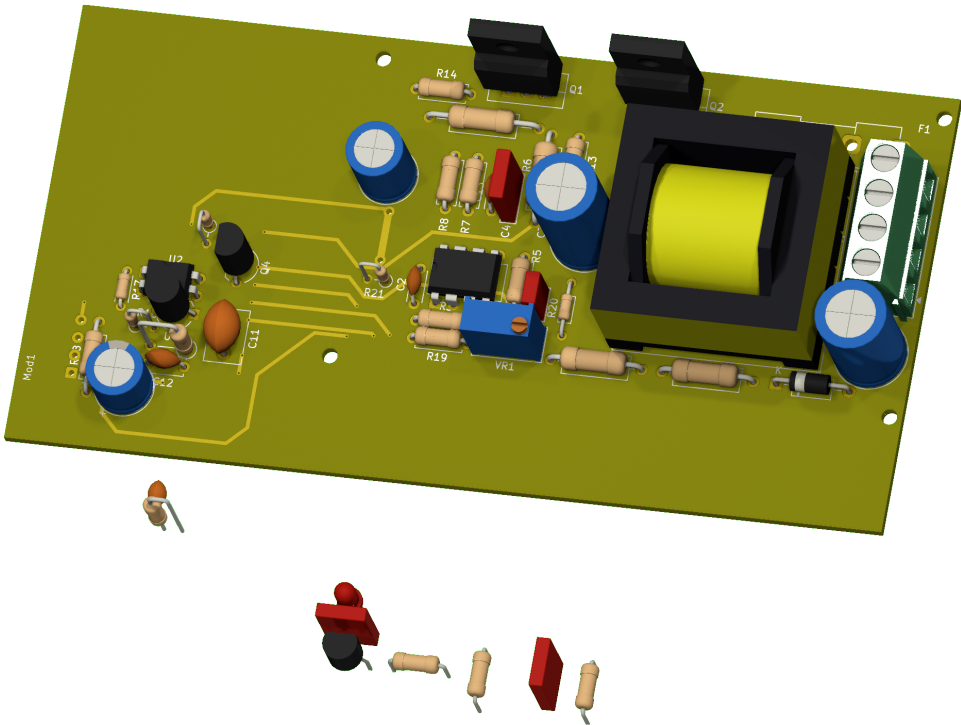
Variant: CHECKED

Rev + (Unreleased)

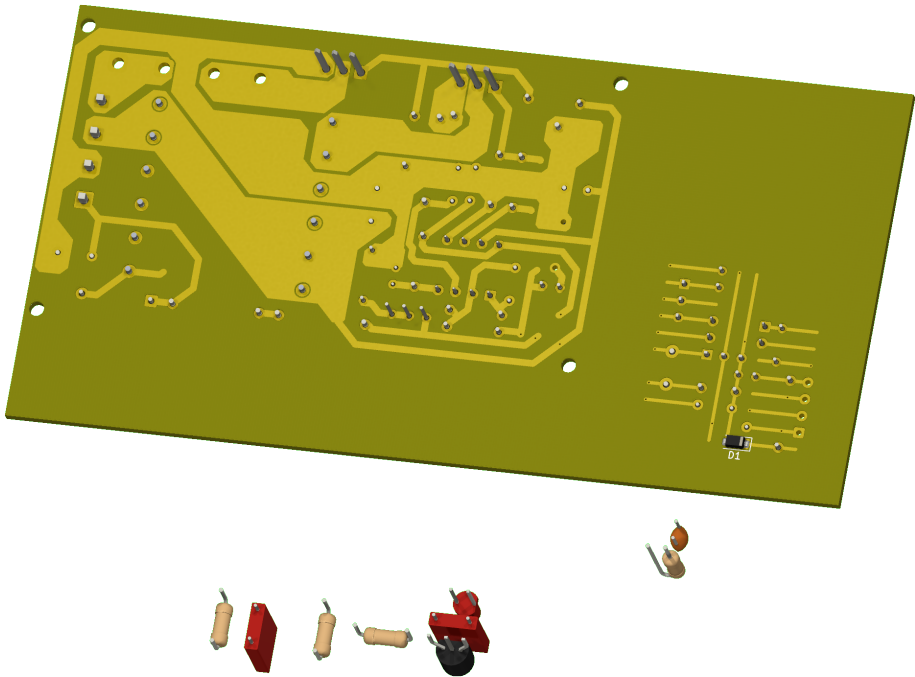
2026-01-03

Page	Index
1	Cover Page
2	Block Diagram
3	Schematic
4	Revision History
5	
6	
7	
8	
9	
10	

TOP VIEW



BOTTOM VIEW



DRAFT - Early concept stage, schematic drafted.

PRELIMINARY - Minor schematic changes are expected, ready for layout planning.

CHECKED - DRC completed, schematic and layout ready for review.

RELEASED - Design ready for manufacturing.

NOTE STYLING CONVENTION

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

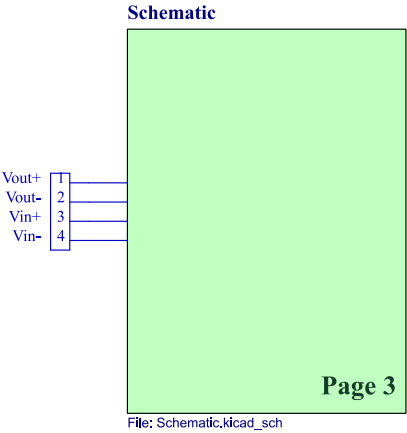
Example text for critical layout guidelines.

Date: 03-Jan-2026

	Comments:	Company: FR	Variant: CHECKED	Git Hash: 120601b
	Sheet Title:	Board Name: Module	Project Name: Isolated step-up module 12-450 V	
	File Name: Isolated_12to450V.kicad_sch	Designer: FR	Date: 2025-12-20	Revision: + (Unreleased)
	Sheet Path: /	Reviewer: NA	Size: A3	Sheet: 1 of 4

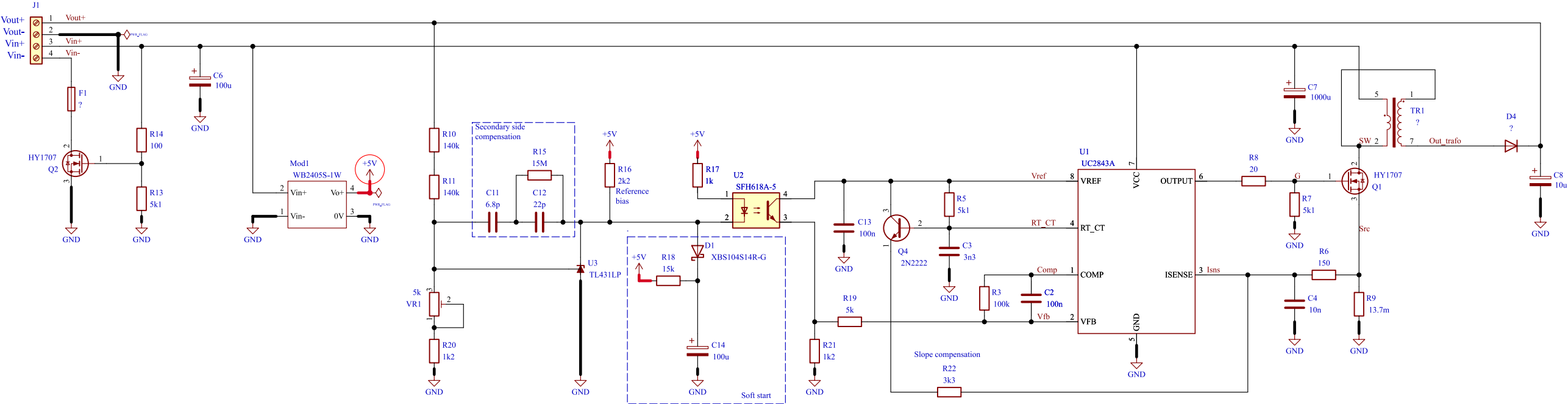
[2] Block Diagram

MODULE INTERFACE:
Vin: 9 V to 24 V according
to AliExpress product page
Vout: 200 V to 450 V,
adjustable through VR1

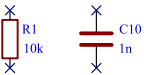
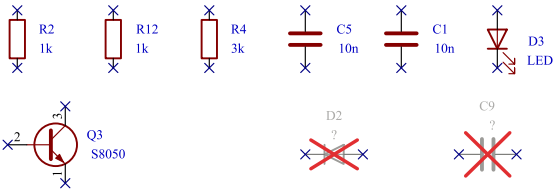


	Comments: AliExpress module example (link)		Company: FR		Variant: CHECKED		Git Hash: 120601b	
			Board Name: Module			Project Name: Isolated step-up module 12:450 V		
	Sheet Title: Block Diagram		File Name: Block Diagram.kicad_sch		Designer: FR		Date: 2025-12-20	
	Revision: + (Unreleased)							
Sheet Path: /Block Diagram/		Reviewer: NA			Size: A4		Sheet: 2 of 4	

[3] Schematic



- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole



Estimated specs for unidentified / custom parts:
- F1: ~5 A
- D4: ~500 mA x 600 V
- R9: ~13.7 mΩ (110 mV @ 8.04 A)
- TR1: ~24:1 turns ratio, 10 μH : 5.76 mH

	Comments:	Company: FR		Variant: CHECKED	Git Hash: 120601b
	Sheet Title: Schematic	Board Name: Module		Project Name: Isolated step-up module 12:450 V	
	Sheet Path: /Block Diagram/Schematic/	File Name: Schematic.kicad_sch	Designer: FR	Date: 2025-12-20	Revision: + (Unreleased)
			Reviewer: NA	Size: A3	Sheet: 3 of 4

[4] Revision History

Version 1.0.0 - 2025-12-19

First issue

Schematic capture complete, except for parts:

- not populated (D2, C9);
- with unreadable or absent markings (F1, D4, R9, TR1).

CI failed on release - do not use.

Version 1.0.1 - 2025-12-21

	Comments:	Company: FR		Variant: CHECKED	Git Hash: 120601b
		Board Name: Module		Project Name: Isolated step-up module 12:450 V	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: FR	Date: 2025-12-20	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer: NA	Size: A4	Sheet: 4 of 4